Units

c	3.00×10^{8} m/s	**1 nm**	10^{-9} m
h	6.63×10^{-34} J·s	**1 MHz**	10^{6} Hz
m_e	9.11×10^{-31} kg	**1 Hz**	1/s
N_A	6.022×10^{23}	**1 amu**	1.661×10^{-27} kg

Trend	→ across period	↓ down group
Z_{eff}	increases	~constant
Atomic radius	decreases	increases
Ionization energy	increases	decreases
Electron affinity	more favorable	less favorable

- Aufbau: 1s 2s 2p 3s 3p **4s 3d** 4p 5s 4d 5p 6s 4f 5d
- Capacity **s2 p6 d10 f14** · orbitals **s1 p3 d5 f7**
- **Exam supplies:** periodic table (Z · group · period · molar mass) and an **EN chart** — read values off them, don't recall them

Watch the wording

"how many **joules** from n grams"	(3) — mol, then × N_A
"longest λ that **ejects** an e^-"	(20) threshold, not (2)
"electron **microscope**", any moving particle	(6)
"falls … **in two steps**"	(5) — add both photon energies
"**unusually big increase**" in IE	(13) — first core e^-
"most **negative** EA" = "**largest** EA"	(14) — same element
"essentially **equal to**" (EA vs IE)	(14) — same magnitude
"contributes **most** to the structure"	(18) — FC nearest 0
"using the **bond energies given**"	(23), not (22)
"**strongest / shortest** bond"	(17) — order before ΔEN

(1) EM Spectrum

	γ	X	UV	vis	IR	μw	radio
Frequency (v)	↑						↓
Energy (E)	↑						↓
Wavelength (λ), nm	<0.01	0.01–10	10–400	400–700	700–10^{6}	10^{6}–10^{9}	>10^{9}

- Visible: violet · blue · green · yellow · orange · red

(2) c = λv (speed of light = wavelength × frequency)

λ × v = cell → divide the cell by whichever you're given

λ ↓ v →	Hz	kHz	MHz	GHz
m	3.00×10^{8}	3.00×10^{5}	3.00×10^{2}	3.00×10^{-1}
nm	3.00×10^{17}	3.00×10^{14}	3.00×10^{11}	3.00×10^{8}

- Check: 500 nm ⇄ 6.00×10^{14} Hz

(3) E = hv = hc/λ (energy = Planck × frequency)

E ↓ given →	λ m	λ nm	v Hz
J / photon	1.99×10^{-25} / λ	1.99×10^{-16} / λ	6.63×10^{-34} × v
kJ / mol	1.20×10^{-4} / λ	1.20×10^{5} / λ	3.99×10^{-13} × v

- kJ/mol row already has ×6.022×10^{23} and ÷1000 built in
- Given grams → mol → ×6.022×10^{23} for photon **count**
- Check with 500 nm: **6.00×10^{14} Hz · 3.98×10^{-19} J · 240 kJ/mol**

(4) Quantum Principles

Planck	Energy emitted only in fixed amounts (quanta)
Einstein	Light arrives in packets (photons) of fixed energy
Heisenberg	Position and velocity not both knowable
de Broglie	Matter has wave properties
Bohr	e^- occupy fixed energy levels
Pauli	No two e^- share all four quantum numbers
Hund	Orbitals fill singly, spins parallel, before pairing

- Orbital = **region where an e^- is most likely found**

(5) Bohr Model & Line Spectra

- e^- occupy **fixed energy levels**; dropping between them emits a photon
- Gives **discrete colored lines**, unique per element — not a continuous band
- Excited H emits in IR, visible, and UV
- Exam list says **no Rydberg**, but HW7 11–12 need the level formula:

$$E_n = -2.18\times10^{-18}\,\mathrm{J}\,(1/n^2)$$
$$\Delta E = -2.18\times10^{-18}\left(\frac{1}{n_f^2}-\frac{1}{n_i^2}\right)\mathrm{J}$$

1. Photon: |ΔE| = hc/λ — go λ ⇄ energy with (3)
2. **Emission** drops ($n_i > n_f$) · **absorption** climbs
3. Falls to ground **in two steps** → add the two photon energies, then solve for n_i

- n = 1 → 4 absorbs $2.18\times10^{-18}(1 - 1/16)$ = **2.04×10^{-18} J**

(6) de Broglie

$$\lambda = h/(mu)$$

1. m → kg: g ×10^{-3}, amu ×1.661×10^{-27}
2. u → m/s: km/s ×10^{3}
3. λ = h/(mu)

- Answer is in m; ×10^{9} for nm

(7) Electron Configuration

1. Fill in Aufbau order (see Units)
2. Respect **s2 p6 d10 f14**
3. Apply the exceptions

- **Cr = [Ar]$4s^1 3d^5$ · Cu = [Ar]$4s^1 3d^{10}$** (half/full d wins)
- Impossible if any subshell is over-filled ($3s^3$, $2p^8$)
- Faster than writing Aufbau: read it off the table — s-block (gp 1–2) → ns, p-block (13–18) → np, d-block → (n−1)d; the period number is n
- Can be a transition metal; ground state only

(8) Reading an Electron Configuration

Paramagnetic vs diamagnetic

1. Draw boxes for the outer subshell — **s 1 · p 3 · d 5 · f 7**
2. Fill singly, parallel spins, before pairing (Hund)
3. Count unpaired e^-

- Any unpaired → **paramagnetic** · all paired → **diamagnetic**
- Full subshells (s^2, p^6, d^{10}) are always paired

Excited vs ground state

- **Ground** = strict Aufbau order, no gaps
- **Excited** = an e^- sits higher than Aufbau requires, leaving a **gap below it**; same total e^- count

Valence electrons

- Count e^- in the **highest n** (outer s + p) = the group number, 1A→1 … 8A→8
- Everything below the highest n is **core**

(9) Ion Configuration

1. Start from the neutral atom
2. **Anion:** add e^- into the outer p until it reaches a noble-gas configuration
3. **Cation:** remove from the **highest n first**

- Transition metals lose **4s before 3d**
- Exam ions are main-group, not transition metals

(10) Penetration, Pauli & Shielding

- **Penetration** — an orbital reaching into the core region is less shielded, so it sits at **lower energy**. Within a shell **s < p < d < f**
- **Pauli exclusion** — no two e^- carry the same four quantum numbers → 2 per orbital, opposite spins
- **Shielding** — only **core** (inner) e^- shield effectively; valence e^- shield each other poorly

(11) Z_{eff} & Core Electrons

$$Z_{eff} \approx Z - (\text{core } e^-)$$

1. Core e^- = total e^- − valence e^- (= the **previous noble gas**)
2. $Z_{eff} \approx Z - \text{core } e^-$

- Same period → same core, so the **rightmost** element wins
- Poor valence shielding is why Z_{eff} rises across a period

(12) Rank Atomic Size

- Down a group **bigger** (new shell) · across a period **smaller** (Z_{eff} pulls in)
- **↑ = lower-left** · **↓ = upper-right**
- Group position beats period position when both apply
- cation < neutral < anion

(13) Rank Ionization Energy

- Across a period ↑ · down a group ↓ (opposite of radius)
- **↑ = upper-right** (He ↑ of all) · **↓ = lower-left**
- $IE_2 > IE_1$ always
- **Big jump** appears once the first **core** e^- is removed

(14) Rank Electron Affinity

- **Most eager for an e^- = halogens, upper right** — this course writes it both as "most negative EA" and "largest positive EA"; either phrasing means the same element
- Noble gases and full/half-full subshells ≈ 0 or unfavorable
- **EA(X) and IE of X^- are the same magnitude** — $F + e^-$ releases exactly what $F^- \rightarrow F$ costs

(15) Families & Diagonal Rule

1A	alkali metals	most reactive metals; ↑ **down**
2A	alkaline earth	reactive, less than 1A
7A	halogens	reactive nonmetals; ↓ **down**
8A	noble gases	**inert** — full octet

- **Diagonal rule** — an element resembles its down-right neighbor: Li~Mg · **Be~Al** · B~Si

HW-only gaps — off the exam list

Oxides — metal oxide = **basic** (K_2O, MgO, Fe_2O_3, Cr_2O_3) · nonmetal oxide = **acidic** (P_4O_{10}, SO_2, CO_2, Cl_2O_7) · Al_2O_3 and BeO = **amphoteric**

Formula from charges — 1A 1+ · 2A 2+ · 3A 3+ · 5A 3− · 6A 2− · 7A 1−; criss-cross and reduce

- Same-group swap keeps the pattern: BaS → **CaO**
- $2A + 3F_2 \rightarrow 2AF_3$ ⇒ A is 3+ (**Al**) · $3Li + Z \rightarrow Li_3Z$ ⇒ Z is 3−, so Mg gives $\mathbf{Mg_3Z_2}$

Bond-type ID — ionic **and** covalent together = a salt of a **polyatomic ion** (NH_4NO_3, K_2SO_4); all-nonmetal = covalent only

Family reactions

- $2K(s) + 2H_2O(l) \rightarrow 2KOH(aq) + H_2(g)$
- $H_2(g) + Cl_2(g) \rightarrow 2HCl(g)$

Odds & ends

- Lanthanides = 4f row (**Ce**) · actinides = 5f (U)
- Diatomic: H_2 N_2 O_2 F_2 Cl_2 Br_2 I_2
- Metallic character ↑ down, ↓ across → **lower-left** wins
- Isoelectronic = matches a noble gas e^- count (K^+, Cl^-, Ca^{2+} = Ar)

(16) Lattice Energy

Energy **released** when gaseous ions form the solid crystal.

$$U \propto \frac{q_1 q_2}{r_1 + r_2}$$

1. Compare charges first
2. If charges tie, smaller ions win

- ↑ charge → ↑ **U** (dominant) · ↑ ionic radius → ↓ U

(17) Electronegativity & Bond Polarity

EN = an atom's ability to attract the electrons shared in a bond.

$$\Delta EN = |EN_A - EN_B|$$

1. Read EN off the chart — EN ↑ up and to the right, F highest, noble gases excluded
2. Take the absolute difference
3. Classify

~0	nonpolar covalent
0.4 – 1.9	polar covalent
≥ 2.0	ionic

- **"Most polar" → ΔEN alone** (largest difference wins)
- **"Strongest / shortest bond" → bond order FIRST:** triple > double > single. ΔEN only breaks ties between equal orders — a C≡N beats a high-ΔEN Si=O
- Larger ΔEN → stronger, shorter bond, but the guide flags this as *not always true*

(18) Formal Charge & Resonance

$$FC = (\text{valence } e^-) - [\text{nonbonding } e^- + \#\,\text{bonds}]$$

1. FC on every atom
2. Check ΣFC = the overall charge
3. Pick the structure with FCs **nearest 0**
4. Put any −1 on the **more electronegative** atom

- Equivalent resonance forms → real bonds are the **average** (e.g. 1½)

(19) Lewis Structures ×4

1. Total valence e^- — **+**1 per negative charge, **−**1 per positive
2. Central atom = least electronegative, never H
3. Single bonds out to every atom
4. Fill outer octets with lone pairs (H takes 2)
5. Leftover e^- onto the central atom
6. Central below 8 → pull an outer lone pair into a double/triple bond, **unless** Al/B/Be
7. Check formal charges — see (18)

Four structures — one of each type:

(a) Normal octet	8 valence e^- per atom (H wants 2) — H_2O, CH_4, NH_3, CO_2, CH_2Cl_2, CH_3CH_2Cl
(b) Incomplete octet	**Al, B, Be only** — content below 8, do not add double bonds to fix. BF_3/BCl_3/AlI_3 leave 6 e^- on the center; $BeCl_2$ leaves 4
(c) Expanded octet	Central atom **period 3 or below** (S, P, Cl, Xe, As, I) uses empty d orbitals for >8 — $AsCl_5$, XeI_2, SO_4^{2-}
(d) Organic	**–OH** alcohol · **C=O** aldehyde/ketone · **–COOH** carboxylic acid · $\mathbf{-NH_2}$ amine

- Every C makes 4 bonds; a –COOH carbon carries one **=O** and one **–O–H**; every O gets 2 lone pairs
- Ions go in brackets with the charge outside
- If the all-single-bond form leaves the center with a nonzero FC, add double bonds to zero it (e.g. two S=O in SO_4^{2-})

(20) Photoelectric Effect

$$KE = h\nu - \Phi = \frac{1}{2}mu^2$$

1. E_{photon} = hν = hc/λ
2. KE = E_{photon} − Φ
3. Speed: u = √(2·KE/m_e)
4. Threshold (KE = 0): λ = hc/Φ

- Φ (also written **W**) = work function = minimum energy to eject an e^-
- Given a threshold λ instead of Φ: Φ = hc/ $\lambda_{threshold}$
- × N_A for a per-mole binding energy
- ↑ **λ** ⇄ ↓ **ν** ⇄ threshold

(21) Heisenberg Uncertainty

$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi}, \quad \Delta p = m\Delta u$$

1. Δp = m·Δu (kg, m/s)
2. Δx ≥ h/(4π·Δp)

- Result is the **minimum** uncertainty
- Rearrange the same way to solve for Δu
- Keep everything in kg, m, s

(22) Born–Haber Cycle

$$\Delta H^\circ{}_f = \Delta H_{sub} + \Sigma IE + \frac{1}{2}BE + EA + U$$

1. List every step with its sign: sublimation (+), IE (+), bond dissociation (+), EA (usually −), U (−)
2. **Halve** the bond dissociation if you need only one atom
3. Chain: elements → gaseous atoms → gaseous ions → solid
4. Set the chain equal to $\Delta H^\circ{}_f$
5. Solve for the single unknown

- Use IE_2 as well when the cation is 2+
- U runs ions(g) → solid, so it comes out **negative**; report |U| if asked for a positive lattice energy

(23) Bond Enthalpy

$$\Delta H_{rxn} = \Sigma\Delta H(\text{bonds broken}) - \Sigma\Delta H(\text{bonds formed})$$

1. List bonds broken (reactants) and formed (products), using the balanced coefficients
2. Σ BE(broken), always positive
3. Σ BE(formed), subtracted
4. ΔH = Σ broken − Σ formed

- Bonds unchanged on both sides cancel — skip them
- Negative ΔH = exothermic
- For "heat released by *g* of X," convert g → mol and scale

Sanity check — expected size

visible photon	~10^{-19} J (a mole ~10^{5} J)
e^- de Broglie λ	~10^{-10} m
photoelectron speed	~10^{5}–10^{6} m/s
Δx (Heisenberg)	~10^{-8} m
lattice energy U	hundreds–thousands kJ/mol
bond enthalpy	150–1000 kJ/mol · ΔH_{rxn} tens–hundreds

- Off by 10^{3} → a unit conversion was skipped. Off by ~6×10^{23} → the mole step.